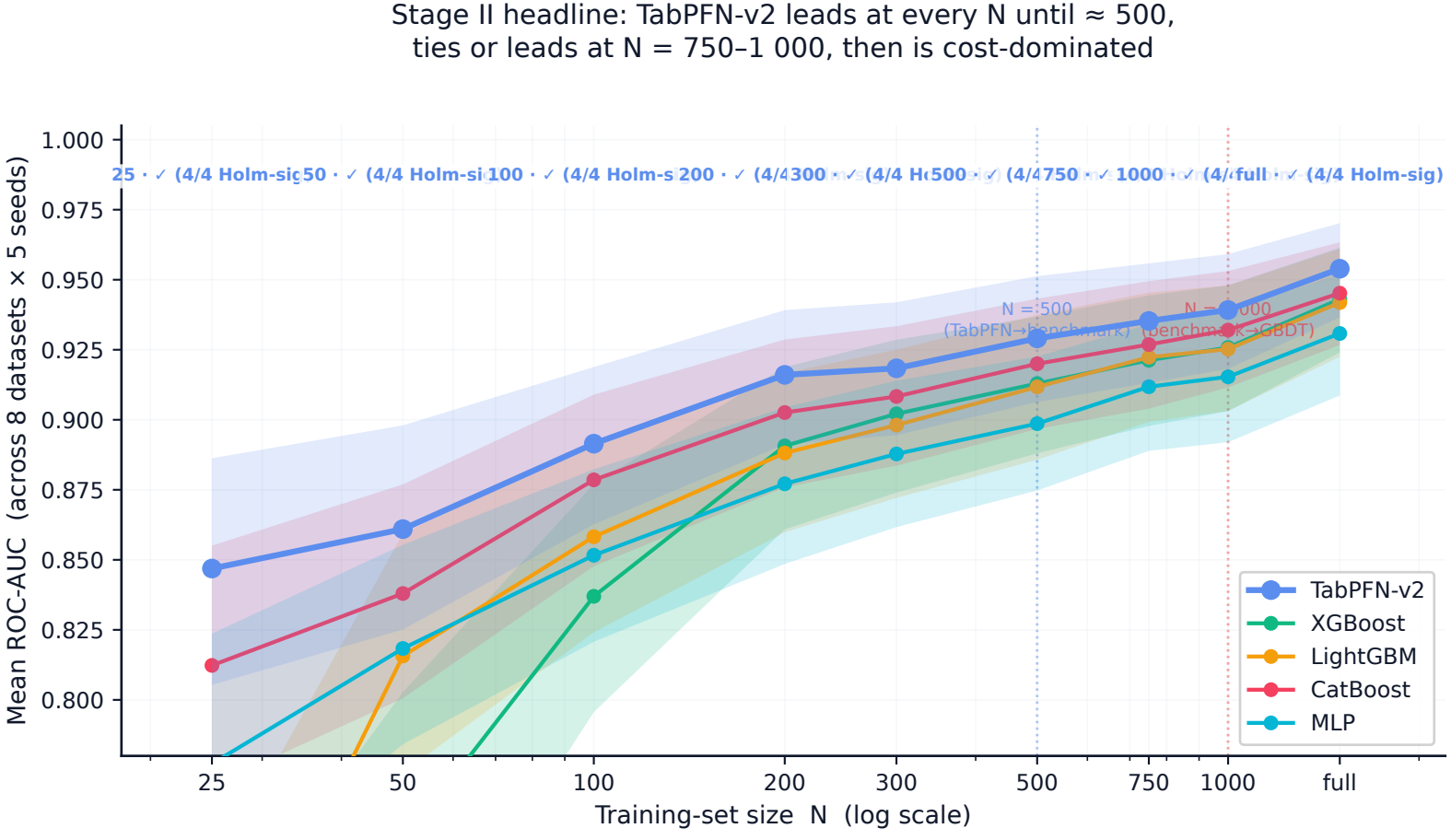

Stage II headline: TabPFN-v2 leads at every N until ≈ 500,
ties or leads at N = 750–1 000, then is cost-dominated
25 · ✓ (4/4 Holm-sig)
50 · ✓ (4/4 Holm-sig)
100 · ✓ (4/4 Holm-sig)
200 · ✓ (4/4
300 · ✓ (4/4 Holm-sig)
500 · ✓ (4/4
750 · ✓ 1000 · ✓ (4/4
full · ✓ (4/4 Holm-sig)
N = 500
(TabPFN→benchmark)
(benchmark→GBDT)
TabPFN-v2
XGBoost
LightGBM
CatBoost
MLP
Mean ROC-AUC (across 8 datasets × 5 seeds)
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.800
25
50
100
200
300
500
750
1000
full
Training-set size N (log scale)